

Cosmological Rupture: The Big Bang as a Physical N_1 Boundary

1. Scope and Intent

This appendix does not propose a new cosmological model, nor does it introduce additional physical assumptions beyond those standard in contemporary cosmology. Its purpose is **interpretative and structural**: to show that the Big Bang can be coherently understood within the present framework as a **physical instance of an N_1 boundary**, rather than as a causal transition between two dynamical regimes.

2. The Big Bang as a Limit of Derivability

In standard cosmology, the Big Bang is not an event embedded within time, but the **lower boundary** of applicability of classical spacetime descriptions. When cosmological equations are extrapolated backward, physical quantities (curvature, density, temperature) diverge, and the governing dynamics cease to be well-defined.

Formally, let the observable universe be modeled as an internally derivable system:

$$N_0^{\text{cosmo}} = (X, T, F),$$

where:

- X is the space of admissible physical states,
- T is cosmological time,
- F encodes local causal dynamics.

Then there exists a boundary time t^* such that:

$$\lim_{t \rightarrow t^*} F \text{ is undefined.}$$

This boundary corresponds to a structural invalidation of derivability, not to a failure of determinism within a valid domain.

3. Planck Scale as Structural Threshold

The Planck scale does not represent a small temporal interval separating two causal chains. Instead, it marks the **limit of admissibility** for the dynamical operator F .

In the language of this framework:

$$S(\omega_t) = 1 \quad \text{for } t > t^*, \quad S(\omega_{t^*}) = 0.$$

Thus, the Planck scale functions as a **structural threshold**, beyond which:

- causal dynamics lose their domain of definition,
 - temporal ordering ceases to be well-defined,
 - and extension by internal derivation becomes impossible.
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4. Big Bang as N_1 , Not as Alternative Dynamics

Crucially, this interpretation rejects the notion that the Big Bang represents a transition between two causal regimes governed by different dynamics F and F' .

If such a transition were derivable, it would require:

- a higher-order dynamic encoding admissibility itself,
- thereby collapsing structure S into a dynamic operator.

This contradicts Theorem 1 (Structure Is Not a Dynamic).

Hence, the Big Bang is not a passage from one chain of causality to another, but the **structural boundary at which the concept of causal chain itself ceases to apply**.

5. Retrodictive Intelligibility

Consistent with Corollary 5, cosmological explanation remains **retrodictive**:

- Physical states for $t > t^*$ are derivable from earlier admissible states.
 - No forward derivation across t^* is possible.
 - The Big Bang is intelligible only as a boundary inferred from within N_0 , not as an origin explained from beyond it.
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6. Structural Interpretation

Within the TNA framework, the Big Bang exemplifies a **physical N_1 event**:

- it invalidates global closure,
- enforces non-derivability across rupture,
- and demonstrates that even fundamental physical theories operate within structurally bounded domains.

This interpretation preserves the empirical success of cosmology while clarifying its intrinsic limits.

7. Summary

The Big Bang is not a causal origin but a **structural limit**.

The Planck scale is not a temporal separator but an **admissibility boundary**.

Cosmology, like all explanatory domains, operates within an N_0 regime bounded by N_1 rupture.

In this sense, the Big Bang provides a canonical physical instance of the structural principles formalized in this paper.

This clarifies why the Big Bang should not be interpreted as a chaotic or stochastic regime, but as a structurally non-derivable boundary where the very conditions of admissible dynamics cease to hold.

Table 1 — Structural Comparison: Big Bang vs Chaos vs Stochastic Breakdown

FEATURE	BIG BANG (STRUCTURAL N_1)	DETERMINISTIC CHAOS	STOCHASTIC BREAKDOWN
Nature of failure	Structural invalidation	Dynamical sensitivity	Probabilistic deviation
Status of dynamics F	Undefined beyond boundary	Fully defined	Fully defined
Domain of applicability	Ends at t^*	Global	Global
Predictability	Impossible beyond boundary	Practically limited, formally intact	Statistically bounded
Role of randomness	None required	None (deterministic)	Essential
Causal structure	Ceases to be admissible	Preserved	Preserved (probabilistically)
Extension across failure	Impossible by derivation	Always possible (but unstable)	Always possible (in distribution)
Explanation after failure	Retrodictive only	Forward + backward (locally)	Statistical inference
Structural operator S	Violated ($S = 0$)	Always satisfied	Always satisfied
Closure of state space	Broken	Preserved	Preserved
Ontological implication	Limit of intelligibility	Epistemic limitation	Noise-driven uncertainty

Interpretive Remark

Chaos and stochastic breakdowns represent **failures of prediction within valid dynamics**. The Big Bang represents a **failure of the dynamics themselves**.

Formally:

- Chaos and stochasticity occur entirely within N_0 .
- The Big Bang marks a transition to N_1 , where derivability is no longer defined.

This distinction is structural, not empirical, and cannot be eliminated by finer measurement, probabilistic modeling, or alternative dynamics without collapsing structure into dynamics—contradicting Theorem 1.

(Theorem 1 <https://zenodo.org/records/18840930>)

Remark — Why the Big Bang Is Not Chaos

It is tempting to interpret the Big Bang as an instance of extreme chaos or stochastic breakdown.

This interpretation is formally incorrect.

Deterministic chaos presupposes the existence of a well-defined dynamical law $F : X \rightarrow X$ acting on a fixed state space, even if trajectories exhibit exponential sensitivity to initial conditions. Stochastic breakdown similarly presupposes a valid probabilistic dynamics defined on the same space.

In contrast, the Big Bang corresponds to a **structural boundary** at which the admissibility conditions themselves fail:

$$S(\omega) = 0$$

Beyond this boundary, no state space X , no time parameter T , and no transition function F remain well-defined.

Thus:

- Chaos is a **failure of predictability within a valid dynamics**.
- The Big Bang is a **failure of the dynamics themselves**.

No refinement of initial conditions, no probabilistic extension, and no alternative parametrization can restore derivability across this boundary without collapsing structure into dynamics—contradicting Theorem 1.

Therefore, the Big Bang is not an instance of chaos or noise, but a paradigmatic case of a physical N_1 event: a rupture of structural admissibility rather than a limit of computation.

Response to Minor Observations

We thank the referee for these insightful remarks. Below we clarify how each point is addressed within the existing framework, and indicate minor additions that strengthen the paper without altering its core claims.

1. Relation to Quantum Gravity

Referee comment:

The paper states that quantum gravity does not resolve the structural problem. This point could be expanded to anticipate the objection that future physics may eliminate the rupture.

Response:

We agree and have clarified this point explicitly. The framework does not deny that quantum gravity may regularize the mathematical singularity (e.g. via a bounce or transition). However, any such resolution necessarily defines a new internally derivable regime N'_0 with its own state space and dynamics F' .

From the perspective of the original regime N_0 , the transition $N_0 \rightarrow N'_0$ remains an N_1 event: the admissibility conditions of N_0 fail, and no internal derivation exists across the boundary. This clarification anticipates and neutralizes the objection that “future physics will solve the problem,” showing that even a successful quantum gravity theory does not eliminate structural non-derivability but merely relocates it.

2. Connection with Entropy and the Arrow of Time

Referee comment:

A connection with entropy and the thermodynamic arrow of time could be mentioned.

Response:

We have added a brief clarification distinguishing these notions. The thermodynamic arrow of time describes irreversibility *within* an admissible regime N_0 , governed by entropy increase under a valid dynamics. By contrast, an N_1 event corresponds to the invalidation of the regime itself.

At the Big Bang, these arrows converge historically but remain conceptually distinct:

- Entropy concerns irreversible evolution inside N_0 .
- Structural rupture concerns the breakdown of N_0 as a domain of derivability.

This distinction prevents conflating structural limits with thermodynamic irreversibility.

3. Status of Inflationary Cosmology

Referee comment:

How does cosmic inflation fit into the N_0/N_1 framework?

Response:

We clarify that inflationary cosmology constitutes an internally derivable regime N_0 in which the Big Bang is dynamically smoothed. However, inflation does not eliminate structural rupture; it displaces it to a pre-inflationary boundary.

Thus, inflation modifies the internal dynamics but leaves the existence of an N_1 boundary intact. This reinforces the claim that N_1 is not a singularity of equations but a limit of admissibility.

Comparative Table — Interpretations of the Big Bang

COSMOLOGICAL FRAMEWORK	BIG BANG DESCRIPTION	STATUS WITHIN REGIME	TNA READING (N0 / N1)
Classical Cosmology (GR)	Initial singularity: density and curvature diverge	Mathematical limit of Einstein's equations	N1: structural rupture, non-derivable
Inflationary Cosmology	Exponential expansion smoothing initial conditions	Valid dynamics within an inflationary N0	N1 displaced to pre-inflation boundary, not removed
Quantum Gravity (Loop, String, Bounce)	Possible transition or bounce avoiding singularity	New regime N0' with its own dynamics	From original N0, still an N1 rupture (non-derivable)
Thermodynamics / Arrow of Time	Minimal entropy at origin, irreversible growth	Irreversibility inside N0	Distinct from N1: entropy ≠ structural invalidation
TNA Framework	Limit of admissibility: dynamics cease to exist	Invalidates global closure and derivability	Canonical physical N1: paradigmatic structural rupture

Editorial Key

- N0 = internally derivable regime with valid dynamics.
- N1 = structural rupture where admissibility fails.
- Any “resolution” (inflation, bounce, quantum regularization) **does not erase the rupture**, it merely **relocates it**.

Contribution in the History of Thought (Clarification)

We appreciate the referee's suggestion to situate the paper historically. The novelty of the present work is not to deny classical positions, but to formalize what they left implicit:

- Aristotle's first cause is reinterpreted as a structural limit, not an efficient cause.
- Kant's cosmological antinomies are resolved by distinguishing temporal beginnings from structural boundaries.
- Hegelian dialectics is contrasted with rupture as formal invalidation rather than synthesis.
- Hawking's claim that "before the Big Bang has no meaning" is given a precise structural explanation.
- Penrose's focus on singularities is reframed in terms of admissibility rather than divergence.

The contribution of this paper is to place these insights within a single axiomatic framework, rather than treating them as isolated philosophical or physical intuitions.

Optional Extension: Experimental Proximity to Structural Limits

We acknowledge the referee's suggestion regarding high-energy physics. While not essential to the present paper, we note that the framework naturally invites questions such as whether increasing energy scales probe regions of configuration space closer to $S(\omega) = 0$. This direction is identified as future work, connecting the formalism to experimental practice without overextending the current scope.

Final Remark

These clarifications do not modify the main results. They sharpen the interpretation and demonstrate that the corollaries and applications follow inevitably from the formal structure introduced. The central claim remains unchanged:

The Big Bang is neither a cause, nor a chaotic regime, nor a stochastic fluctuation, but a structurally non-derivable boundary.